GANTA BHANU GANESH

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.58
Narayana Junior College, Visakhapatnam	Class XII (Board of Intermediate Education, Andhra Pradesh)	2015	95.40%
Minerva Public School, Visakhapatnam	Class X (Board of Secondary Education, Andhra Pradesh)	2013	9.70%

SKILLS

Programming Languages: Python, MATLAB, C, HTML, CSS, JavaScript.

Libraries and Frameworks: Pandas, NumPy, Matplotlib, Seaborn, TensorFlow, PyTorch, Scipy, Scikit-learn.

Simulation Tools: MATLAB Simulink, FASTSim, LTSpice, HOMER, PVsyst, FASTSim.

Hardware platforms: Raspberry Pi, ESP 8266 Node MCU, Arduino, ESP32

Soft skills: Leadership, Problem-solving abilities, Collaboration and Teamwork, Mentorship, Time Management, Adaptability, Critical Thinking, Self-motivation.

Languages known: English, Telugu, Hindi.

PATENTS

- "**Enhanced Powertrain Architecture for Electric Vehicle Applications considering the Add-on Battery**", **Granted Patent**, Indian Patent number: 583993. Inventors: **Bhanu Ganesh Ganta** and Pradeep Kumar Yemula.
- "**A Fault Protection System For A Dual Battery Electric Vehicle**" **Published Patent**, Indian Patent Application No. 202541134197. Inventors: **Bhanu Ganesh Ganta** and Pradeep Kumar Yemula.
- "**Fault-Responsive Isolation and Failover Operation in a Dual-Battery Electric Vehicle**" **Published Patent**, Indian Patent Application No. 202541134719. Inventors: **Bhanu Ganesh Ganta** and Pradeep Kumar Yemula.

PROJECTS

An Interaction-aware multi-physics framework for EV energy estimation under real-world conditions.

- Developed a **physics-based, interaction-aware mathematical model for EV energy estimation**, considering **57 parameters and their coupled interactions** among multiple subsystems using Python.
- Proposed a **closed-loop framework** that updates **6 state variables** to capture cross-domain energy dynamics.
- Validated the framework using **real-world driving data** from a **158 km round-trip** on a 4-wheeler passenger EV along Hyderabad Outer Ring Road (ORR), achieving a final state-of-charge (SOC) estimation **error of 3.92%**.
- Performed **subsystem deactivation analysis**, showing that **excluding HVAC overestimated SOC by 14.25%**, while **excluding BTMS overestimated SOC by 16.6%**, increased battery temperature from 30.96°C to 33.04°C, and reduced energy consumption by 2.11 kWh.

Feasibility Study of Integrating an Add-On Battery (AOB) for Electric Vehicle Applications.

- Analyzed the **technical feasibility** of integrating an AOB for EVs based on energy consumption, driving range,

charging time.
- **Developed a mathematical model** to evaluate EV energy consumption across different battery capacities.
- A **6.43% increase in vehicle weight** due to the AOB increased rolling resistance, gradient, and acceleration forces by **5.70%, 6.44%, and 5.76%**, respectively.
- An EV with an AOB **reduces travel time by 55 min** for a **400km trip** while **increasing energy consumption by 2.86 kWh** and operating cost by Rs. 15 compared with an EV without an AOB.
- An EV equipped with an AOB can cover a 400 km distance in the same time while providing cost **savings of Rs. 1020 and Rs. 1800 over diesel and petrol vehicles**, respectively.
- Performed a **techno-economic analysis** considering battery replacement and operational costs, demonstrating a minimum lifetime profit of **Rs. 40 lakh** for an ABCSS operator with an initial battery inventory of 20.

Parameter Perturbation Analysis of EV Energy Consumption

- Performed **parameter perturbation analysis** by varying one parameter at a time while maintaining all other parameters constant to quantify their individual impact on energy consumption.
- Analyzed 8 key parameters by **individually perturbing each across 10 scenarios** (five above and below actual values) using 115 km of real-world driving data, and **proposed a sensitivity index** to rank their relative impact.

Vehicle Control Architecture for Dual-Battery Electric Vehicles with Multi-Scenario Protection

- Developed a **Stateflow-based fault-tolerant control** architecture enabling **seamless failover** in a dual-battery EV.
- Proposed **fail-safe fault-protection strategy** enables seamless failover to the healthy main battery by electrically and mechanically isolating a faulty add-on battery during **under-voltage, over-current, and internal short-circuit** faults.
- Developed **UML sequence diagrams** to model subsystem interactions during faults and battery transition.
- Demonstrated **reliable operation**, improving fault tolerance, operational continuity, system reliability, and user safety.

Enhanced Powertrain Architecture for Electric Vehicles considering an Add-On Battery

- Designed a **novel dual-battery powertrain architecture** by integrating a **removable add-on battery (AOB)** with the primary traction battery, enabling extended driving range and enhanced operational flexibility.
- Proposed a **safe battery changeover mechanism** that permits switching from the main battery (MB) to the AOB only when the **MB load current is zero** and the pressure sensor confirms the **mechanical presence of the AOB**.
- Enabled independent battery operation, eliminating current-sharing control and the associated power losses.

Master's Project: Optimal Sizing of PV/Wind/Battery Stand-alone and Grid-connected System Using TLBO.

- **Formulated the optimization problem** by incorporating technical and operational constraints, including Loss of Load Probability (LOLP), battery State of Charge (SOC), and system capacity limits.
- Developed **mathematical models** of PV, wind, battery energy storage, and load for determining HRES performance.
- **Optimized PV array, wind turbine, and battery storage capacities** to minimize LCOE while maintaining LOLP <= 5%, comparing six metaheuristic algorithms in MATLAB for optimal system sizing at a selected Indian location.
- Conducted **techno-economic analysis** evaluating system reliability, RE utilization, unmet load, and lifecycle cost.

Real-time Campus Energy Monitoring System (CEMS)

- Developed **CEMS** for real-time acquisition of **29** electrical parameters using a **Raspberry Pi** and RS-485 communication.
- Implemented a Python-based automated data acquisition system to retrieve meter data at 1 min intervals, enabling simultaneous monitoring of **40+ energy meters**.
- Designed **web-based SCADA** dashboards for real-time visualization, analytics, and operational decision-making.
- Deployed the CEMS across **5+ client sites**, supporting both single-phase and three-phase energy meters.
- Implemented automated email alerts for abnormal operating conditions, including **low power factor and 90% CMD**.

IoT-Enabled 1-phase AC Smart Meter Prototype

- Developed an **IoT-enabled single-phase AC smart meter prototype** for real-time monitoring of voltage, current, power, energy, frequency, and power factor with a 10s sampling interval, **rated for 230 V and 100 A**.
- Integrated the **PZEM-004T sensing module** with an **ESP8266 microcontroller** for real-time data acquisition and implemented Wi-Fi-based communication with Google Apps Script for cloud data logging in Google Sheets.

IoT-Enabled DC Smart Meter Prototype

- Developed an **IoT-enabled DC smart meter prototype** for real-time monitoring of battery discharge characteristics using the **PZEM-017** sensing module and an **ESP8266 microcontroller**.
- Fetches voltage, current, power, and energy every 10 s, designed for measurements up to 300 V and 300 A, and implemented Wi-Fi-based communication via Google Apps Script for data logging in Google Sheets.

PUBLICATIONS

- **B.G.Ganta**, P.K.Yemula, D.S.Pawar and A.Boruah, "Assessment of Residual Range of an Electric Vehicle: Experimental Validation using Real World Driving Cycles", IEEE IECON 2024, USA, doi: IECON55916.2024.
- **B.G.Ganta**, R.S.Nalluri, V.S.B.Chaitanya, and P.K.Yemula, "Assessing Impact of Dynamic Parameters on Electric Vehicle Energy Consumption using Parameter Perturbation Analysis", IEEE ICPS 2025, India, doi: ICPS.4978.
- **B.G.Ganta**, B.K.Reddy, and P.K.Yemula, "Real-time Monitoring of Energy Dynamics in Electric Vehicles", IEEE ICPS 2025, India, doi: ICPS.4876.
- **B.G.Ganta**, H.P.Golla, P.K.Yemula, and B.K.Reddy "The Role of Distribution Utilities in Enabling Electric Vehicle Ecosystems: A Multi-Perspective Analysis with Case Studies", IEEE ITEC AP 2025, Singapore, doi: ITECAP5001.
- **B.G.Ganta**, H.P.Golla, P.K.Yemula, and B.K.Reddy "Sustainability in Electric Transportation Systems: Concepts and Case-Based Analysis", IEEE ITEC AP 2025, Singapore, doi: ITECAP4854.
- K.S.Middya, **B.G.Ganta**, and P.K.Yemula, "Fast Charging Techniques for Lithium-Ion Batteries: A Comparative Analysis and Algorithmic Innovations for Safety and Degradation Mitigation", IEEE ITEC INDIA 2025, India, doi: ITEC9337.
- A.Bhimaraju, **G. Bhanu Ganesh**, and A. Mahesh, "Optimal sizing of PV/wind/battery grid-connected hybrid renewable energy systems using TLBO algorithm," Springer Nature 2024, Singapore, doi: Springer9235.
- A.Bhimaraju, **G.B.Ganesh**, A.Mahesh, Optimal sizing of PV/ Wind/ Battery standalone hybrid renewable energy systems using TLBO algorithm, Springer 2023, Singapore, doi: Springer9285.
- V.S.B.Chaitanya, **G.B.Ganesh**, P.K.Yemula et al., "Real-Time Energy Data Extraction and Analysis of a Nanogrid System in BVRIT Campus for Critical Loads Using IoT Modules", IEEE ISGT Asia 2024, India, doi: ISGT6250.
- V.S.B.Chaitanya, Josna, **B.G.Ganta**, and P.K.Yemula, "Impact of ToD Charges and Power Factor on Total Energy Costs of BVRIT Campus Distribution Grid," IEEE ICPS 2025, India, doi: ICPS4986.

SCHOLASTIC ACHIEVEMENTS

- Received the **Silver Award** for **Best Student Paper** titled "**Feasibility Study and Economic Analysis of an Add-on Battery for Electric Vehicles**" at the International Conference on Sustainable Energy (ICUE), 2024.

ACADEMIC OUTREACH AND PROFESSIONAL CONTRIBUTIONS

- **Delivered a seminar** titled "**Electric Vehicle Charging Infrastructure for Green Mobility**" as part of the Student Seminar Series at Rajiv Gandhi University of Knowledge Technologies, Nuzvid (2022).
- Contributed as a **Reviewer** for **IEEE Conferences** (INDICON 2024, ISGT Asia 2024, ICPS 2025, and SPERT 2025), reviewing manuscripts related to electric mobility, power systems, and storage technologies.
- **Delivered** multiple sessions on "**EV Deployment - Global and Indian Scenario, Technological Developments in the EV Domain, and EV Charging Infrastructure**" to DISCOM engineers at ESCI, Hyderabad.

POSITIONS OF RESPONSIBILITY

PhD Mentor in Sunshine, the counseling cell at IIT Hyderabad (IITH).

PG- PhD Management Head in Sunshine, the counseling cell at IIT Hyderabad.

Inspection Coordinator, Mess Monitoring Committee in the student mess at IITH.

EXTRACURRICULAR

- **Travel & Hospitality Coordinator** for the **ICPS 2025 Conference**, managing travel logistics for invited speakers.

- **Finance coordinator** for the **MSME Tech Connect 2026**, managing the event budget and bills processing.